Lecture 9

Covariance and Correlation

20.04.2026

Outline

Expected value of $g(X, Y)$

Recap: expectation of random variable $g(X)$

- Discrete case $E[g(X)] = \sum_{\text{all } x} g(x)f(x)$
- Continuous case $E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$

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Suppose $g(X, Y)$ is a real-valued function of random variables X and Y, then

- Discrete case

$$E[g(X,Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x,y)f(x,y)$$

- Continuous case

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)f(x,y)dxdy$$

Example: let X and Y be random variables with joint pdf $f(x, y)$. Find $E(X + Y)$

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- This can be generalized to n random variables

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$$

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$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n)$$

- How about $E(XY)$?

Let X and Y be INDEPENDENT random variables with joint pdf $f(x, y)$. Find $E(XY)$.

Let X and Y be INDEPENDENT random variables with joint pdf $f(x, y)$. Find $E(XY)$.

$$\begin{aligned}
E(XY) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf(x, y)\ dx\ dy \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_X(x)f_Y(y)\ dx\ dy \\
&= \int_{-\infty}^{\infty} yf_Y(y)\left[\int_{-\infty}^{\infty} xf_X(x)dx\right]dy \\
&= E(X)\int_{-\infty}^{\infty} yf_Y(y)dy \\
&= E(X)E(Y)
\end{aligned}$$

Note: (1) this formula only holds when X and Y are independent.
(2) This is not a sufficient condition for independence.

Exercise

What's the expected value of $X - Y$?

Outline

Covariance

Definition

Covariance of two random variables X and Y is defined as

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- Simplification

$$\begin{aligned} Cov(X,Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY + \mu_X \mu_Y - X\mu_Y - Y\mu_X]) \\ &= E[XY] - \mu_X \mu_Y \end{aligned}$$

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- Recall

$$\begin{aligned} E[XY] &= \int \int xy \; f(x,y) \; dx \; dy \quad \text{if continuous} \\ &= \sum_x \sum_y xy \; f(x,y) \quad \text{if discrete} \end{aligned}$$

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- $Cov(X+a, Y+b) = Cov(X,Y)$

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Covariance of sums of random variables

$$Cov\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} Cov\left(X_i, Y_j\right)$$

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A special case

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2\sum_{1\leq i<j\leq n} Cov(X_i, X_j)$$

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Some more special cases

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$
$$Var(X-Y) = Var(X) + Var(Y) - 2Cov(X,Y)$$

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$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i)$$

- $Cov(X,Y)=0 \not\Longleftarrow X$ and Y are independent
  Counter example?

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Covariance

$$E[X] = 0,\ E[XY] = E[X^3] = \int_{-0.5}^{0.5} x^3 dx = \frac{x^4}{4}\Big|_{-0.5}^{0.5} = 0$$

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Independence:

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Independence: since Y depends on X, so not independent.

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(To be more rigorous, we need to show

$$f(x, y) \neq f_X(x) f_Y(y)$$

for some $x, y \in \mathbb{R}$.)

Correlation

Since $Cov(X, Y)$ depends on the magnitude of X and Y we would prefer to have a measure of association that is not effected by arbitrary changes in the scales of the random variables.

Definition

The most common measure of <u>*linear*</u> *association is correlation which is defined as*

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)}\,\sqrt{Var(Y)}}$$

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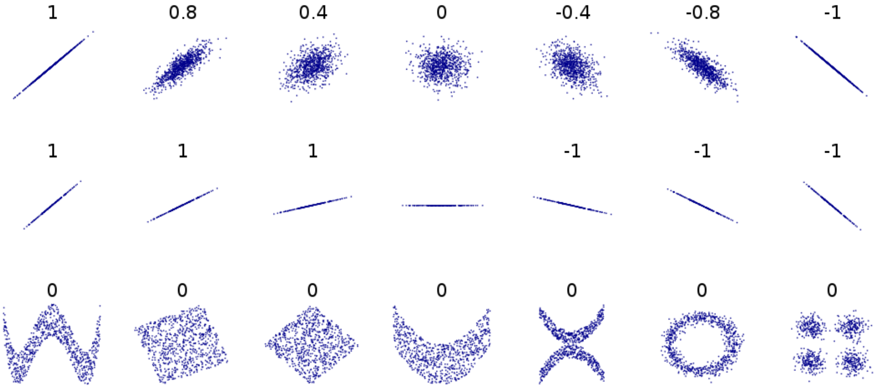
Definition

The most common measure of linear association is correlation which is defined as

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)}\,\sqrt{Var(Y)}}$$

- range: $-1 \leq \rho(X, Y) \leq 1$
- the magnitude (i.e. absolute value) of the $\rho(X, Y)$ measures the strength of the linear association
- the sign determines if it is a positive or negative relationship.
- if $\rho(X, Y) = 0$, then X and Y are said to be uncorrelated.

Correlation



Exercise

Suppose that a bivariate r.v. (X, Y) is uniformly distributed over a unit circle.

a. Are X and Y independent?
b. Are X and Y correlated?

Exercise

The joint pdf of (X, Y) is given by

$$f_{X,Y}(x,y) = \begin{cases} xye^{-(x^2+y^2)/2}, & x > 0, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

a. Find the marginal pdf's of X and Y.
b. Are X and Y independent?

Outline

Conditional expectation

- The discrete case: for all $p_Y(y) > 0$
  - Conditional pmf: $p_{X|Y}(x \mid y) = P\{X = x \mid Y = y\} = \frac{p(x,y)}{p_Y(y)}$

Definition

The conditional expectation of X given that $Y = y$ is

$$E[X \mid Y = y] = \sum_x xP\{X = x \mid Y = y\} = \sum_x xp_{X|Y}(x \mid y)$$

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- The continuous case: for all $f_Y(y) > 0$
  - Conditional pdf: $f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}$

Definition

The conditional expectation of X given that $Y = y$ is

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} xf_{X|Y}(x \mid y)dx$$

Properties of expectations remain

- Expectation of a function of a random variable
  - The discrete case:

$$E[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x \mid y)$$

  - The continuous case:

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- Expectation of sum of random variables

$$E[\sum_{i=1}^{n} X_i \mid Y = y] = \sum_{i=1}^{n} E[X_i \mid Y = y]$$

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$$E[\sum_{i=1}^{n} X_i \mid Y = y] = \sum_{i=1}^{n} E[X_i \mid Y = y]$$

- Think about the condition $Y = y$ as taking expectation of X on a reduced sample space consisting only of outcomes for which $Y = y$

Computing expectations by conditioning

$$E[X] = E[E[X \mid Y]]$$

- A very important property of conditional expectation
- Think of $E[X|Y]$ as a random variable (when $Y = y$)
- The discrete case:

$$E[X] = \sum_{y} E[X \mid Y = y] P\{Y = y\}$$

- Intuition:
  - $E[E[X \mid Y]]$ is a weighted average of $E[X \mid Y]$, where weights are $P\{Y = y\}$ (the probability of the condition)
  - Similar to the "law of total probability" $P(E) = \sum_{i=1}^{n} P(E \mid F_i)P(F_i)$

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A miner is trapped in a mine containing 3 doors. The 1st door leads to a tunnel that will take him to safety after 3 hours of travel. The 2nd door leads to a tunnel that will return him to the mine after 5 hours of travel. The 3rd door leads to a tunnel that will return him to the mine after 7 hours. If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

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- Let X denote the amount of time in hours until he reaches safety, and Y denote the door he initially chooses.

$$\begin{aligned} E[X] &= E[X \mid Y=1]P\{Y=1\} + E[X \mid Y=2]P\{Y=2\} + E[X \mid Y=3]P\{Y=3\} \\ &= \frac{1}{3}(E[X \mid Y=1] + E[X \mid Y=2] + E[X \mid Y=3]) \end{aligned}$$

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- Note that $E[X|Y=1] = 3, E[X \mid Y=2] = 5 + E[X], E[X \mid Y=3] = 7 + E[X]$
- Therefore $E[X] = \frac{1}{3}(3 + 5 + E[X] + 7 + E[X])$, which gives $E[X] = 15$

Computing probabilities by conditioning

- Let E denote an arbitrary event, and define the indicator random variable X as
$$X = \begin{cases} 1 & \text{if } E \text{ occurs} \\ 0 & \text{if } E \text{ does not occur} \end{cases}$$
- Then $E[X] = P(E), E[X \mid Y = y] = P(E \mid Y = y)$ for any random variable Y
- The discrete case:
$$P(E) = \sum_y P(E \mid Y = y) p(Y = y)$$
related to $P(E) = \sum_{i=1}^n P(E \mid F_i) P(F_i)$

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Conditional variance

- Similarly to the conditional expectation, we can define the conditional variance of X given that $Y = y$

Definition

$$Var(X \mid Y = y) = E[(X - E[X \mid Y = y])^2 \mid Y = y]$$

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- A very useful conditional variance formula

$$Var(X) = E[Var(X \mid Y)] + Var(E[X \mid Y])$$